Synthesis and Planning:
Arsenic Colorimeter Methods

Design notes

April 22, 2026

Where we are

You have a hardware-first project with a working optical prototype, a clear measurement goal (arsenic in drinking water, with the decision threshold around the WHO 10 µg/L limit), a chemistry you don't fully control (gas-phase Gutzeit-type strip reaction), and a statistics pipeline that was written somewhat ahead of the hardware and the data. The original `.tex` is a speculative methods document — a plausible skeleton assembled from standard tools (Mahalanobis, inverse-Wishart, GP), not a reflection of measurements you've actually looked at.

My critique flagged two kinds of issues: mathematical consistency problems inside the skeleton, and physical difficulties the skeleton didn't acknowledge. Your response splits those cleanly:

- **Physical issues you've already thought about and have a plan for:** illumination non-uniformity (polynomial flat-field), strip heterogeneity (defer, study empirically), strip kinetics (built-in timer, fixed protocol), temperature (ambient + water + LED-adjacent probes), stray light (flat-black interior, small patch area, testable).
- **Physical issues still open:** batch-to-batch strip variation, the exact spectral behavior of the IMX219 under each LED, HDR / dynamic range in the pale-yellow regime, dark current behavior vs. exposure time and temperature.
- **Statistical issues you're explicitly deferring to collaboration:** what to actually use as features, what to regress with, what to report for uncertainty, and how to validate.

My job at this planning stage is to (a) design a statistical pipeline that fits your hardware plan, (b) tell you what data I need to see before committing to specifics, and (c) flag what you should learn so you're not flying blind on the statistics.

The approach I'd recommend, at the level of strategy

The original paper's architecture was: *within-image Mahalanobis distances to reference patches, stacked with LED temperatures, fed to a GP.* I don't think that's the right shape. Here is the shape I'd recommend instead, with the reasoning.

Build the pipeline in layers, each with a clear job, each testable on its own. The layers are: raw capture → pixel-level corrections → patch-level feature extraction → dimensional reduction → regression/classification → uncertainty calibration. Each layer should be swappable.

The temptation with a project like this is to combine layers — e.g., fold flat-fielding into the covariance model — and that makes debugging impossible.

Layer 1: Raw capture and pixel-level corrections

This is the part where your hardware plan already lives. Concretely:

- **Dark subtraction first, always.** Every image, raw, minus a same-exposure-time dark frame. Dark-subtract before anything else. The dark frame captures read noise bias, dark current, and fixed-pattern noise in one shot. Your plan mentions "subtract the mean black background" but a *mean* dark throws away pattern noise. Subtract the full dark *frame*, pixel-by-pixel. Capture darks periodically (they drift with sensor temperature).
- **Flat-field correction using your polynomial plan.** Your plan is sound. One refinement: fit the polynomial *per Bayer channel* (you already said this) and *per LED* (you already said this), but also consider fitting across a wider area than just the rectangles and evaluating at rectangle centers. A polynomial fit over more pixels is better conditioned. Degree-2 is almost certainly right; only go to degree-3 if residuals show structure.
- **Saturation and dead-pixel masking.** Before any mean, mask pixels that are saturated ($\geq \sim$1000 DN on a 10-bit sensor) or identified as hot/dead from calibration. The IMX219 has $\sim$0.1% bad pixels typically. Patch means should be trimmed means or medians over unmasked pixels.
- **Exposure bracketing.** I'd strongly consider capturing each LED condition at two or three exposures and selecting (or HDR-combining) to maximize SNR without saturating. The pale-yellow regime is where you need the dynamic range.

The output of this layer is: per-patch, per-LED, per-Bayer-channel, a flat-fielded, dark-subtracted, outlier-masked pixel array. Not a mean yet — keep the pixels around, because the spatial distribution of values inside the strip patch is itself information (see Layer 3).

Layer 2: Within-image normalization

This is where your existing plan of "divide by white patch" lives, but I'd do it differently than the original `.tex`.

- After flat-fielding, the white reference patch should read approximately constant across the frame (that's what flat-fielding is for). So *any* white patch on the card, or an average of several, gives you a single scalar per (LED, channel) to divide by. This is the illumination-intensity normalizer.
- The black patch, after dark subtraction, ideally reads zero. If it doesn't, you have stray light, and the black patch value is your stray-light estimate. Subtract it. (This is your stray-light test, built into every measurement.)
- The result is a *reflectance-like* quantity per patch, per LED, per channel: dimensionless, between 0 and 1, interpretable as "fraction of incident light reflected."

This is a much cleaner foundation than the original paper's setup. It cleanly separates intensity normalization (white) from stray-light subtraction (black). Both are physical quantities, not statistical nuisances.

Layer 3: Feature extraction from patch pixels

Here I'd depart from the original paper significantly. The paper jumped straight to "patch mean, then Mahalanobis distance to references, with pixel-noise covariance in the middle." I'd reconsider.

What you actually have from Layer 2 is, per strip, a multivariate reflectance value per (LED, channel) position — roughly $I \times 4$ values if all Bayer channels are used for all LEDs, minus the ones knocked out by low SNR. For typical cases that's maybe 8–12 numbers.

The useful features are:

- **Mean reflectance vector.** The obvious one. Call it $\mathbf{r} \in \mathbb{R}^N$ where N is the number of (LED, channel) combinations surviving SNR culling.
- **Within-patch variability.** The standard deviation or inter-quartile range of pixel reflectance inside the strip patch, per (LED, channel). This is a *uniformity flag* — if the strip reacted unevenly, these values will be elevated compared to calibration. Not a predictor of concentration, but a quality indicator.
- **Ratios / log-ratios between channels.** For a colorimetric measurement, ratios are often more informative than absolute values. A log-reflectance vector $-\log(\mathbf{r})$ is the right transform if you believe Beer-Lambert-like behavior (which is an approximation for strips, but a useful one).

I would **not** compute Mahalanobis distances to reference patches as the primary feature. Here's why: reference patches are useful for *rebasing the measurement against known colors present in the same image*, which corrects for illumination drift. But that's already handled by white-patch normalization in Layer 2. The references give you a *check* — does a "yellow reference" in this image read the same as it did in calibration? — but they aren't the input to the regression. The regression input should be the strip's reflectance vector itself.

Keep the reference patches in the pipeline, but in a different role: **as a per-image quality gate.** If any reference patch's reflectance vector in the live image differs from its calibration mean by more than k standard deviations (this *is* where a Mahalanobis distance earns its keep — as an outlier detector on known-color patches), flag the image and recalibrate or reject.

Layer 4: Dimensional reduction and covariate assembly

You anticipated this: PCA on the strip reflectance vectors across a calibration set will likely reveal that 1–2 components capture $> 95\%$ of the variance, because there's essentially one chemical axis (white→yellow→brown). That's the "principal color score" s_1, and possibly s_2.

The feature vector for regression is then small:

$$\mathbf{x} = \big(s_1,\ s_2,\ T_{\text{ambient}},\ T_{\text{water}},\ t_{\text{develop}},\ \text{strip-lot ID}\big).$$

Low-dimensional, interpretable, and each component has a physical meaning. This is *vastly* preferable to the original paper's $IJ + I$-dimensional feature vector for a GP.

Layer 5: Regression

Here the question is not "GP vs. not GP" but "what's the simplest model that works?" In order of increasing complexity:

1. **Univariate calibration curve.** $c = f(s_1)$. Fit a monotone spline, or a parametric curve motivated by the chemistry. This should be your baseline and it will probably cover 80% of cases. It's intelligible to engineers. Every colorimetric analysis textbook uses this.
2. **Bivariate.** $c = f(s_1, s_2)$. Include s_2 if residuals from (1) show structure along s_2.
3. **With covariates.** Add T_{water}, t_{develop}, etc. if residuals correlate with them.
4. **GP, only if needed.** Switch to a GP if a parametric model doesn't fit well and you have enough calibration data to justify it. GPs are powerful but opaque. Save them for when you've demonstrated that simpler models aren't sufficient.

The original paper's "GP with a 12+ dim feature vector" is almost certainly overfit for a calibration set of a few hundred strips.

Layer 6: Uncertainty calibration and decision threshold

This is absent from the original and matters enormously for a field instrument. Two things:

- **Empirical prediction intervals via cross-validation or conformal prediction.** Rather than trust the model's internal uncertainty (GP posterior variance, or fitted spline confidence bands), compute residuals from leave-one-strip-out CV, and use their distribution to build honest 95% prediction intervals. Report those.
- **Decision-relevant metrics.** The WHO limit is 10 µg/L. What matters is the *false-negative rate at 10 µg/L*. Design the validation around this specifically. ROC-like analysis centered on the decision threshold is more important than overall RMSE.

What I need to know to commit to specifics

Before I'd write down equations or code, I need the following, in roughly this order of importance:

1. **The actual strip chemistry.** Is this a Gutzeit ($HgBr_2$-on-paper) strip? Arsenator? Something else? The kinetics, temperature sensitivity, and spectral signature differ. This determines whether we expect Beer-Lambert-like log-reflectance behavior, what the dominant color axis looks like, and whether there's one reaction product or several.
2. **A small real dataset.** Even 20–50 strips across the concentration range, imaged on the prototype, would let me see the actual shape of the color manifold, the noise level, and whether 1-D PCA really is enough. Without this we're just speculating.
3. **The LED choices and their spectra.** You said "blue, green, red, and optionally more." Which exact LEDs? Peak wavelengths, FWHMs? More LEDs aren't automatically better if their spectral bands don't align with features in the strip's reflectance spectrum.
4. **The reference card design.** How many references, what colors? Are they printed (potentially fugitive), Munsell chips, or Pantone-painted? Long-term stability of the reference card is critical for an instrument that will live in the field.
5. **Target performance.** LOD? Required precision at 10 µg/L? Required precision at higher concentrations? Time budget per measurement? This drives how much calibration effort is justified.

6. **Deployment constraints.** How much computation can a Raspberry Pi Zero do per measurement? A full GP inverse is fine for a few hundred points; a cross-validated conformal prediction interval means doing that several times. This bounds what's implementable.

What I recommend you learn or read

Not huge detours, but enough to let you judge proposals rather than accept them:

- **Beer-Lambert and diffuse reflectance basics.** Why log-reflectance is often the right transform. Kubelka-Munk is the standard model for reflectance from scattering media like paper strips; it's a one-page read and gives you a language for discussing what's going on optically. This is the single most useful thing to read.
- **PCA intuition, not mechanics.** You already have the mechanics from the `.tex`. What's worth building up is the intuition that PCA on a calibration set tells you the *dimensionality of the chemistry*, and the explained-variance plot is a diagnostic tool you'll use to argue (to yourself and to reviewers) that a 1-D or 2-D model is justified.
- **Cross-validation and why "training error" lies.** Hastie, Tibshirani, Friedman (*Elements of Statistical Learning*) chapter 7 is the classic. The key idea: you will always be able to fit your calibration data well; the honest question is how well you predict *unseen* strips, and CV is the answer.
- **Conformal prediction, at a high level.** This is a method for building honest prediction intervals around *any* regression model without distributional assumptions. A short intro paper suffices. For a field instrument where the end user will make a health decision from the number, this matters.
- **What Mahalanobis distance actually is, and where it does and doesn't belong.** Not a deep dive — the intuition "distance measured in units of standard deviations, accounting for correlation between dimensions" is enough, plus the fact that it requires you to specify *which* covariance. In a classifier it's the within-class covariance. In outlier detection it's the expected-under-the-null covariance. In the original paper it was within-patch pixel noise, which was the wrong choice.

What the rewritten document should look like

When we're ready to write, I'd structure it as:

1. Instrument and measurement concept (hardware, chemistry, protocol, timing).
2. Image processing pipeline (Layers 1–3 above) — this is the engineering meat.
3. Feature extraction and calibration model (Layers 4–5), presenting the univariate model as the baseline and more complex models as extensions justified by residual analysis.
4. Uncertainty quantification and decision thresholds (Layer 6).
5. Validation protocol and results (once data exists).

Substantially shorter and substantially less mathematical than the original. The original's weakness was that it paraded statistical machinery; the new version's strength should be that each step has a specific, defensible reason rooted in the physics and chemistry.

Immediate next step I'd suggest

Get any real data — even a quick prototype pass with ten strips at five concentrations, imaged with whatever LEDs you have — and look at it with me. Everything above is informed speculation. A plot of $\log(1/r)$ vs. concentration for each LED, over a real run, answers more questions in ten minutes than another round of theory.